

Production functional realisation – scoped T5 promotion enactment

TECT verification-first repository · claim A1-PRODUCTION-FUNCTIONAL-REALISATION

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1. Purpose and scope

This note enacts the scoped T5 promotion of the manifest-defined, standalone three-component production-functional backend. The closed object is the hash-pinned discrete spectral-torus implementation on $N = 4, 6, 8$, with the listed five field classes, finite-difference steps, parameters, and shell-bias activation. It is not a repair or validation of the separately hash-pinned historical working branch, which remains a recorded variational audit failure. No continuum, PDE, minimizer, BCC-selection, or stability result is asserted.

2. Frozen functional and implementation

For $\rho = \Psi^\dagger \Psi$, $m_A = \Psi^\dagger T_A \Psi$, $J_A = \nabla m_A$, and $K_A = J_A - (m_A/(\rho + 10^{-12}))\nabla \rho$, the backend implements

$$F = F_{\text{Braz}} + F_{\text{family}} + F_{\text{lock}} + F_{\text{shell}} + \int \left(\frac{\lambda}{4} \rho^2 + \frac{\gamma}{6} \rho^3 \right) + \sum_A \int \left(\frac{a}{2} |J_A|^2 + b \operatorname{Re}(\bar{J}_A \cdot K_A) + \frac{c}{2} |K_A|^2 \right),$$

where $a = c_{JJ} \alpha_X^2 / (M_X^2 + 10^{-12})$, $b = c_{JK} \alpha_X \beta_X / (M_X^2 + 10^{-12})$, and $c = c_{KK} \beta_X^2 / (M_X^2 + 10^{-12})$. The manifest fixes the Brazovskii term, family mass, internal lock, shell bias, quartic and sextic terms, all three Class-II coefficients, density regularizers, projector, and real pairing convention. The backend, verifier, and collector are SHA-256 pinned.

3. Closed verification matrix

The separately constructed verifier checks

$$DF[\Psi](v) = \langle R[\Psi], v \rangle_{\mathbb{R}}.$$

$$DR[\Psi]v = H[\Psi]v.$$

$$\langle u, Hv \rangle_{\mathbb{R}} = \langle Hu, v \rangle_{\mathbb{R}}.$$

It independently defines field generation, the real torus pairing, centred energy and residual differences, and the analytic scalar reduction. It covers zero, homogeneous, random, q_0 -shell, and Class-II-active fields at the pinned point and at $\eta_{\text{shell}} = 0.07$, for $N = 4, 6, 8$ and steps $10^{-4}, 10^{-5}, 10^{-6}$.

The persisted maximum relative errors are

$$1.856661591692621 \times 10^{-7}, \quad 6.695217715947473 \times 10^{-9}$$

$$1.282693395943113 \times 10^{-10}, \quad 6.218564371786818 \times 10^{-16}$$

for the variational identity, $DR = H$, Hessian symmetry, and scalar reduction, respectively. All source, hash, version, grid, field-coverage, and coupling activation assertions pass.

4. Independent reproduction and confirmation

The operator-named evidence package [runs/promotion-evidence/2026-07-17-jusang-independent/](#) records reviewer Jusang Lee, the frozen source hashes, the command, a clean subprocess result, and REPRODUCTION-PASS. Its six evidence-file hashes also verify. On 2026-07-17 the operator requested verification of that result and approval of the scoped T5 promotion. This approval applies only to this note’s discrete implementation scope and authorises the final claim-level publication bundle.

5. Devil’s-advocate

(alpha) “*The Class-II cross term could be omitted.*” DISMISSED: the manifest and nonzero activation check require distinct c_{JJ} , c_{JK} , and c_{KK} coefficients.

(beta) “*Autodiff makes the test circular.*” VALID with mitigation: the verifier independently evaluates energy directional differences, residual differences, the real pairing, and the scalar reduction.

(gamma) “*This proves a continuum functional.*” UPHELD as an invalid reading: only the explicitly enumerated discrete grids and tests are closed.

(delta) “*This validates the old solver or identifies a BCC vacuum.*” UPHELD as an invalid reading: the old source remains failed, and neither solver integration nor BCC/minimizer/stability evidence is part of this result.

(epsilon) “*T5 is a theorem claim.*” UPHELD as an invalid reading: T5 is a hash-pinned, reproducible closure within its stated scope, not T6 or T7.

6. Result footer

Result ID:	A1-PRODUCTION-FUNCTIONAL-REALISATION
Precise statement:	The hash-pinned standalone all-coupling backend closes the declared discrete variational test matrix.
Scope:	Spectral torus N=4,6,8; five field classes; three steps; pinned parameters plus declared shell activation only.
Dependencies:	A1-PRODUCTION-KERNEL-MANIFEST.
Evidence grade:	EXECUTED, CONDITIONAL; independent REPRODUCTION-PASS.
Reproduction command:	python codes/foundations/a1_production_backend_verify.py --grids 4 6 8
Expected output:	PRODUCTION-BACKEND-MULTIGRID-PASS; all assertions PASS.
Falsification gate:	hash/version drift or any failed required assertion.
Tier before / after:	T4 / T5. CLOSED@DISCRETE-VARIATIONAL-MATRIX.
No-overclaim statement:	No old-solver repair, continuum/PDE, minimizer, BCC, stability, T6, or T7 claim.
Next required action:	Use only at this pinned T5 scope; create separate claims for solver integration, convergence, or physical selection.